

Assignment 3: Data Exploration

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Exploration.

Directions

1. Rename this file `<FirstLast>_A03_DataExploration.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Assign a useful **name to each code chunk** and include ample **comments** with your code.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.
7. After Knitting, submit the completed exercise (PDF file) to the dropbox in Canvas.

TIP: If your code extends past the page when knit, tidy your code by manually inserting line breaks.

TIP: If your code fails to knit, check that no `install.packages()` or `View()` commands exist in your code.

Set up your R session

1. Load necessary packages (tidyverse, lubridate, here), check your current working directory and upload two datasets: the ECOTOX neonicotinoid dataset (ECOTOX_Neonicotinoids_Insects_raw.csv) and the Niwot Ridge NEON dataset for litter and woody debris (NEON_NIWO_Litter_massdata_2018-08_raw.csv). Name these datasets “Neonics” and “Litter”, respectively. Be sure to include the sub-command to read strings in as factors.

```
#Workspace setup  
#Load necessary packages  
install.packages(c("tidyverse", "lubridate", "here"))  
library(tidyverse); library(lubridate); library(here)  
here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```

#Load in datasets
#Start with "Neonics"
Neonics <- read.csv(
  file = here("Data", "Raw", "ECOTOX_Neonicotinoids_Insects_raw.csv"),
  stringsAsFactors = TRUE
)

#Now do "Litter"
Litter <- read.csv(
  file = here("Data", "Raw", "NEON_NIWO_Litter_massdata_2018-08_raw.csv"),
  stringsAsFactors = TRUE
)

```

Learn about your system

2. The neonicotinoid dataset was collected from the Environmental Protection Agency's ECOTOX Knowledgebase, a database for ecotoxicology research. Neonicotinoids are a class of insecticides used widely in agriculture. The dataset that has been pulled includes all studies published on insects. Why might we be interested in the ecotoxicology of neonicotinoids on insects? Feel free to do a brief internet search if you feel you need more background information.

Answer: We might be interested in the efficacy of this type of insecticide on insects. Alternatively, we could be interested in the effects that insecticides could have on other non-pest insects that might encounter it in the environment from aerial dispersion or runoff.

3. The Niwot Ridge litter and woody debris dataset was collected from the National Ecological Observatory Network, which collectively includes 81 aquatic and terrestrial sites across 20 ecoclimatic domains. 32 of these sites sample forest litter and woody debris, and we will focus on the Niwot Ridge long-term ecological research (LTER) station in Colorado. Why might we be interested in studying litter and woody debris that falls to the ground in forests? Feel free to do a brief internet search if you feel you need more background information.

Answer: Litter and woody debris is a valuable part of nutrient cycling in terrestrial systems. We might be interested in the amount of litter in a system and how that might affect the microbial or decomposer systems and contribute to larger system function.

4. How is litter and woody debris sampled as part of the NEON network? Read the [NEON_Litterfall_UserGuide.pdf](#) document to learn more. List three pieces of salient information about the sampling methods here:

Answer: 1. Litter and woody debris are caught in elevated and ground traps. 2. Mass from each collection event is measure and categorized. 3. The traps are placed in plots based on the vegetation, so they can be random or targeted locations.

Obtain basic summaries of your data (Neonics)

5. What are the dimensions of the dataset?

```

#Data Dimensions
#Check the dimensions of the dataset
dim(Neonics)

```

```
## [1] 4623 30
```

6. Using the `summary` function on the “Effect” column, determine the most common effects that are studied. Why might these effects specifically be of interest? [Tip: The `sort()` command is useful for listing the values in order of magnitude...]

```
#Data Summary of Effects  
#Summarize the values in the "effect" field and sort by magnitude  
sort(summary(Neonics$Effect))
```

```
##      Hormone(s)      Histology      Physiology      Cell(s)  
##           1           5           7           9  
##      Biochemistry      Accumulation      Intoxication      Immunological  
##           11          12          12          16  
##      Morphology      Growth      Enzyme(s)      Genetics  
##           22          38          62          82  
##      Avoidance      Development      Reproduction      Feeding behavior  
##          102          136          197          255  
##      Behavior      Mortality      Population  
##          360          1493          1803
```

Answer: The most common effects are population (1803), mortality (1493), and behavior (360). These are significant measures of toxicological effects and can show the efficacy of the insecticides.

7. Using the `summary` function, determine the six most commonly studied species in the dataset (common name). What do these species have in common, and why might they be of interest over other insects? Feel free to do a brief internet search for more information if needed.[TIP: Explore the help on the `summary()` function, in particular the `maxsum` argument...]

```
#Data Summary of Species  
#Determine the most frequently studied species  
summary(Neonics$Species.Common.Name, maxsum = 7)
```

```
##      Honey Bee      Parasitic Wasp Buff Tailed Bumblebee  
##           667           285           183  
##      Carniolan Honey Bee      Bumble Bee      Italian Honeybee  
##           152           140           113  
##      (Other)  
##          3083
```

Answer: Honey bee, Parasitic wasp, Buff tailed bumblebee, Carniolan honey bee, Bumblebee, and Italian honeybees are the most studied. This is because bees are an essential pollinator, so researchers are likely interested in determining the impacts of the insecticides on non-target species, such as pollinators.

8. Concentrations are always a numeric value. What is the class of `Conc.1..Author.` column in the dataset, and why is it not numeric? [Tip: Viewing the dataframe may be helpful...]

```
#Data Concentrations  
#look at field class  
class(Neonics$Conc.1..Author.)
```

```
## [1] "factor"
```

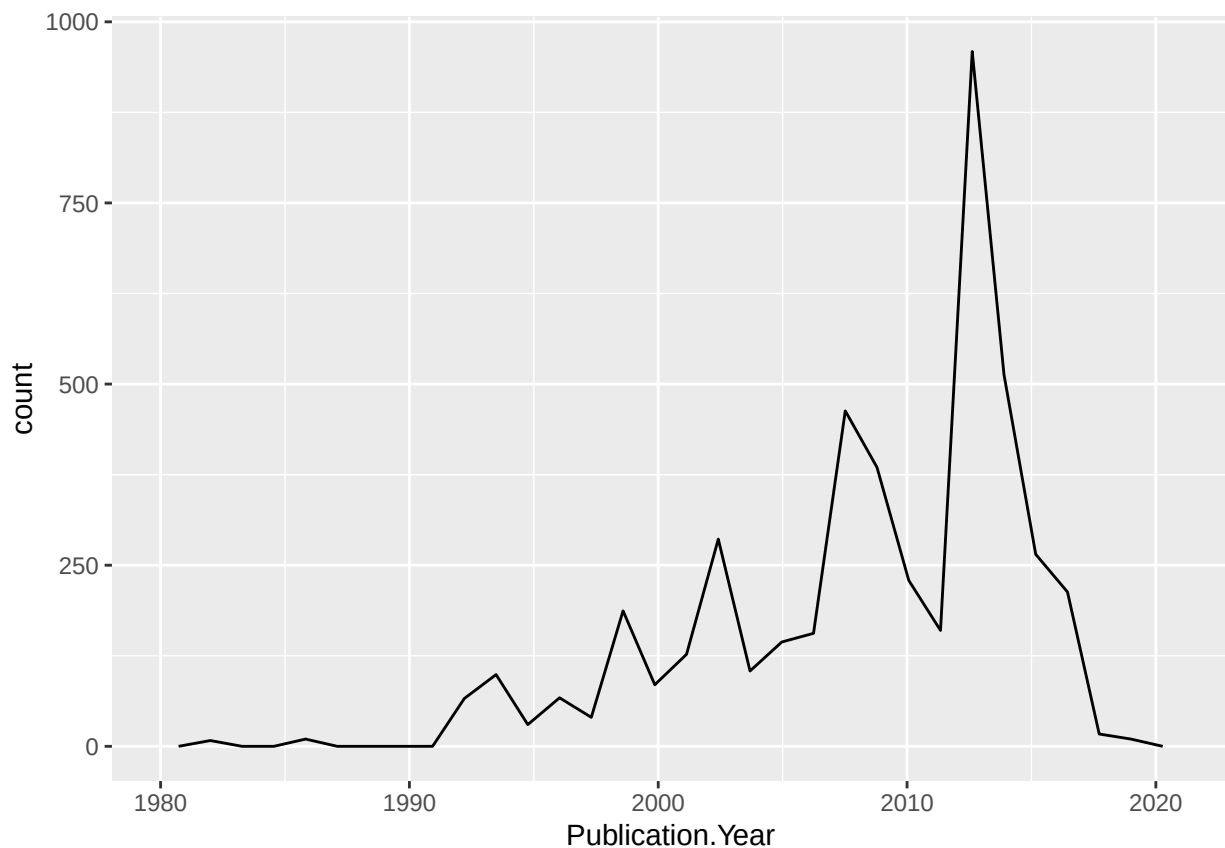
Answer: The Conc.1..Author field is a factor class. This is because there is not a consistent data type, so the data would have been read as a string, but we opted to read strings as factors.

Explore your data graphically (Neonics)

9. Using `geom_freqpoly`, generate a plot of the number of studies conducted by publication year.

```
#Graph Neonics in a Frequency Polygon  
#use ggplot to plot the number of publications by year, using a bin width of 30  
ggplot(Neonics, aes(x=Publication.Year)) +  
  geom_freqpoly()
```

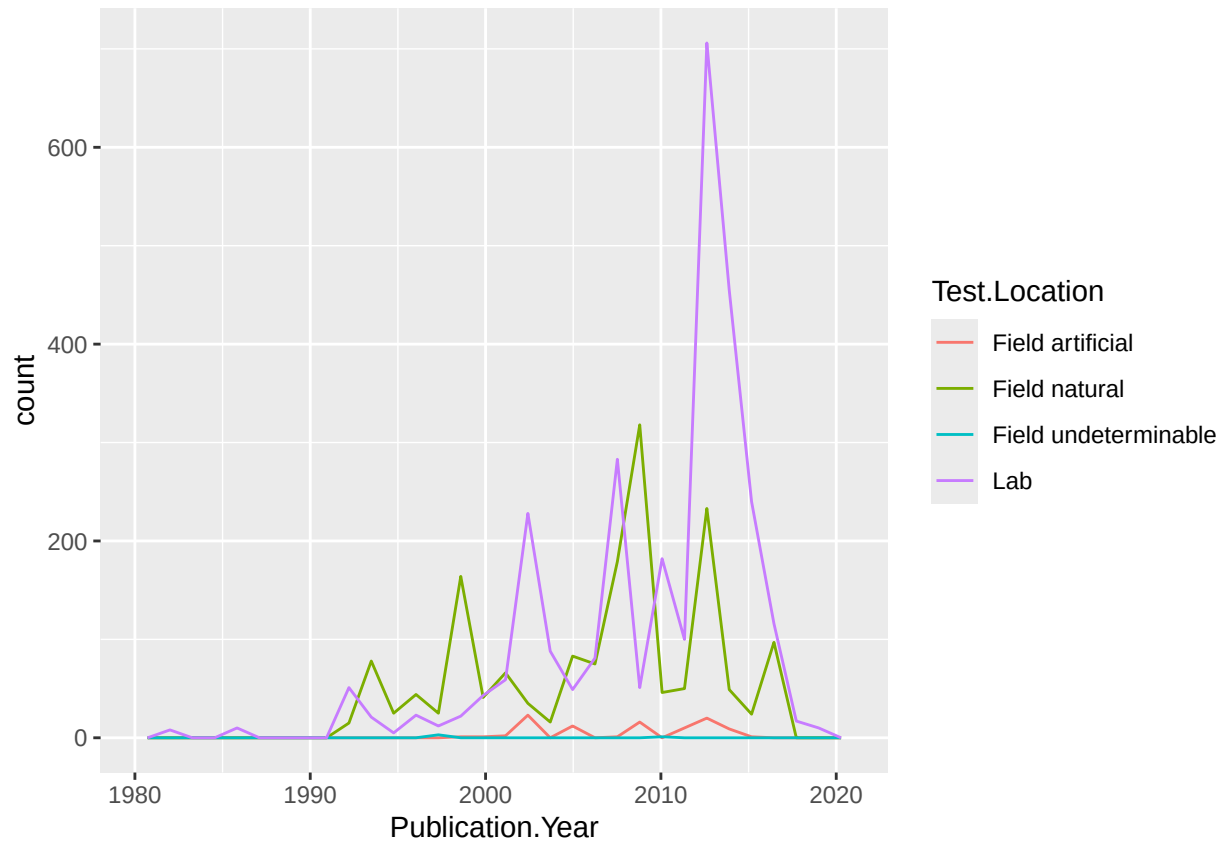
```
## 'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.
```



10. Reproduce the same graph but now add a color aesthetic so that different Test.Location are displayed as different colors.

```
#Graph Neonics in a Frequency Polygon  
#use ggplot to plot the number of publications by year, using a bin width of 30 and coloring bars by location  
ggplot(Neonics, aes(x=Publication.Year, colour=Test.Location)) +  
  geom_freqpoly()
```

'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.



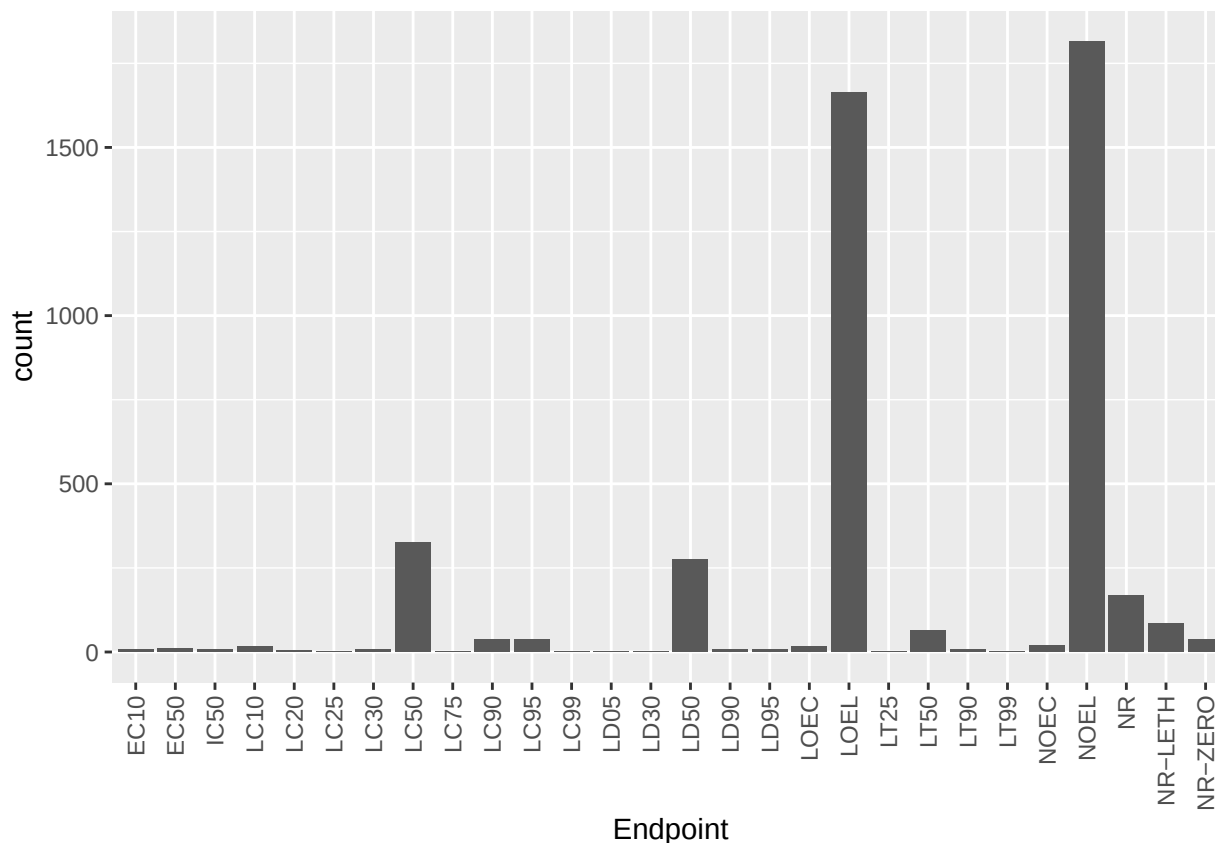
Interpret this graph. What are the most common test locations, and do they differ over time?

Answer: The most common location is in a lab setting, but before around 2012, natural field studies and lab studies were occurring with similar frequency. Then, lab studies massively spiked in the mid-2010s.

11. Create a bar graph of Endpoint counts. What are the two most common end points, and how are they defined? Consult the ECOTOX_CodeAppendix for more information.

[**TIP:** Add `theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))` to the end of your plot command to rotate and align the X-axis labels...]

```
#Plot Endpoints
#use ggplot to plot the frequency of endpoints.
ggplot(Neonics, aes(x=Endpoint)) +
  geom_bar() +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1))
```



Answer: NOEL (no observable effects level) and LOEL (Lowest observable effect level) are the most common endpoints. NOEL represents the highest concentration of a contaminant that is not observable or not significantly different from responses to controls. LOEL is lowest concentration at which response will be observably different than control treatment response.

Explore your data (Litter)

- Determine the class of collectDate. Is it a date? If not, change to a date and confirm the new class of the variable. Using the `unique` function, determine which dates litter was sampled in August 2018.

```
#Field Exploration - collectDate
#investigate the field class of collection dates and ensure it is in date format
class(Litter$collectDate)
```

```
## [1] "factor"
```

```
Litter$collectDate <- as.Date(Litter$collectDate)
unique(Litter$collectDate)
```

```
## [1] "2018-08-02" "2018-08-30"
```

```
#Data was collected August 2nd and 30th
```

13. Using the `unique` function, list the different plotIDs sampled at Niwot Ridge. How is the information obtained from `unique` different from that obtained from `summary`?

```
#Unique Plots  
#first, check to make sure there are no other sites  
unique(Litter$siteID)
```

```
## [1] NIWO  
## Levels: NIWO
```

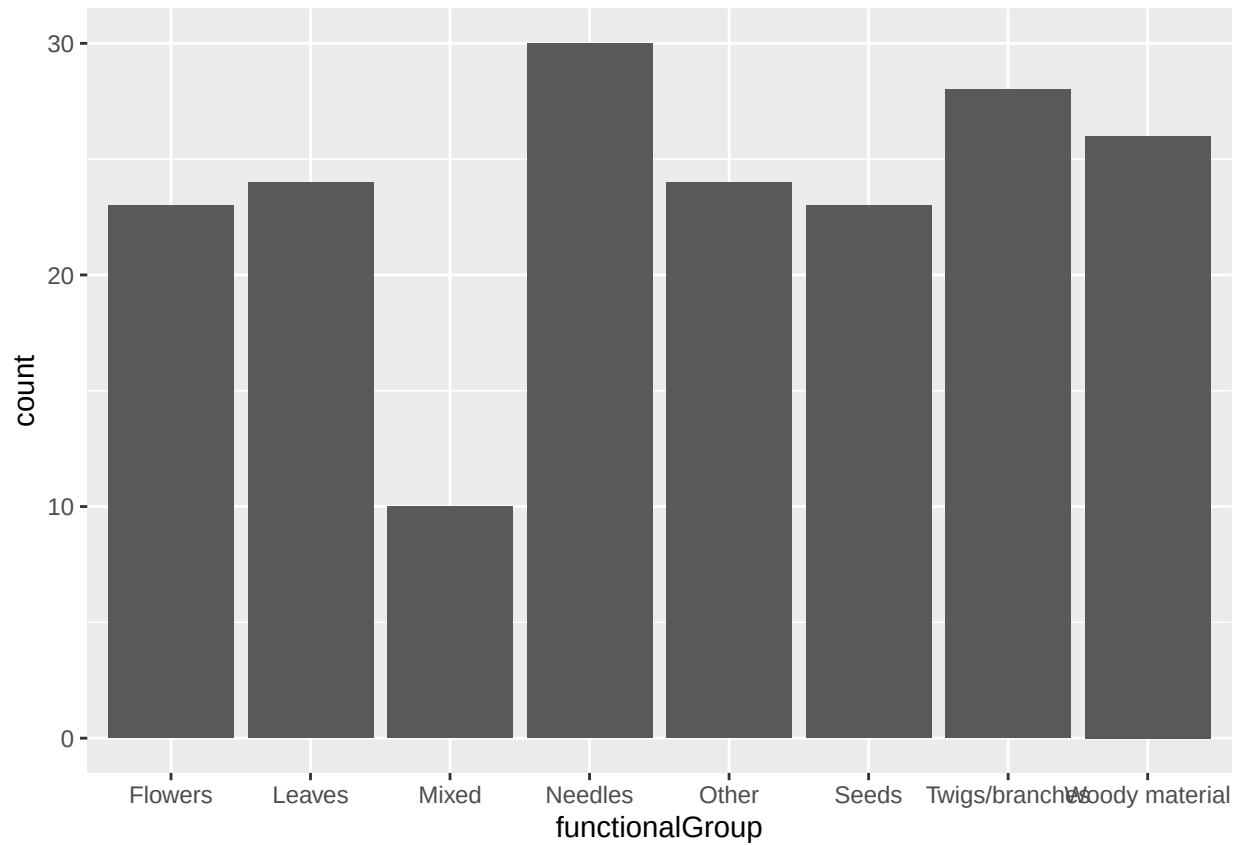
```
#then, check the number of plot IDs  
unique(Litter$plotID)
```

```
## [1] NIWO_061 NIWO_064 NIWO_067 NIWO_040 NIWO_041 NIWO_063 NIWO_047 NIWO_051  
## [9] NIWO_058 NIWO_046 NIWO_062 NIWO_057  
## 12 Levels: NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 ... NIWO_067
```

Answer: Unique lists all of the unique IDs, while `summary` gives a count of how many times each ID appears.

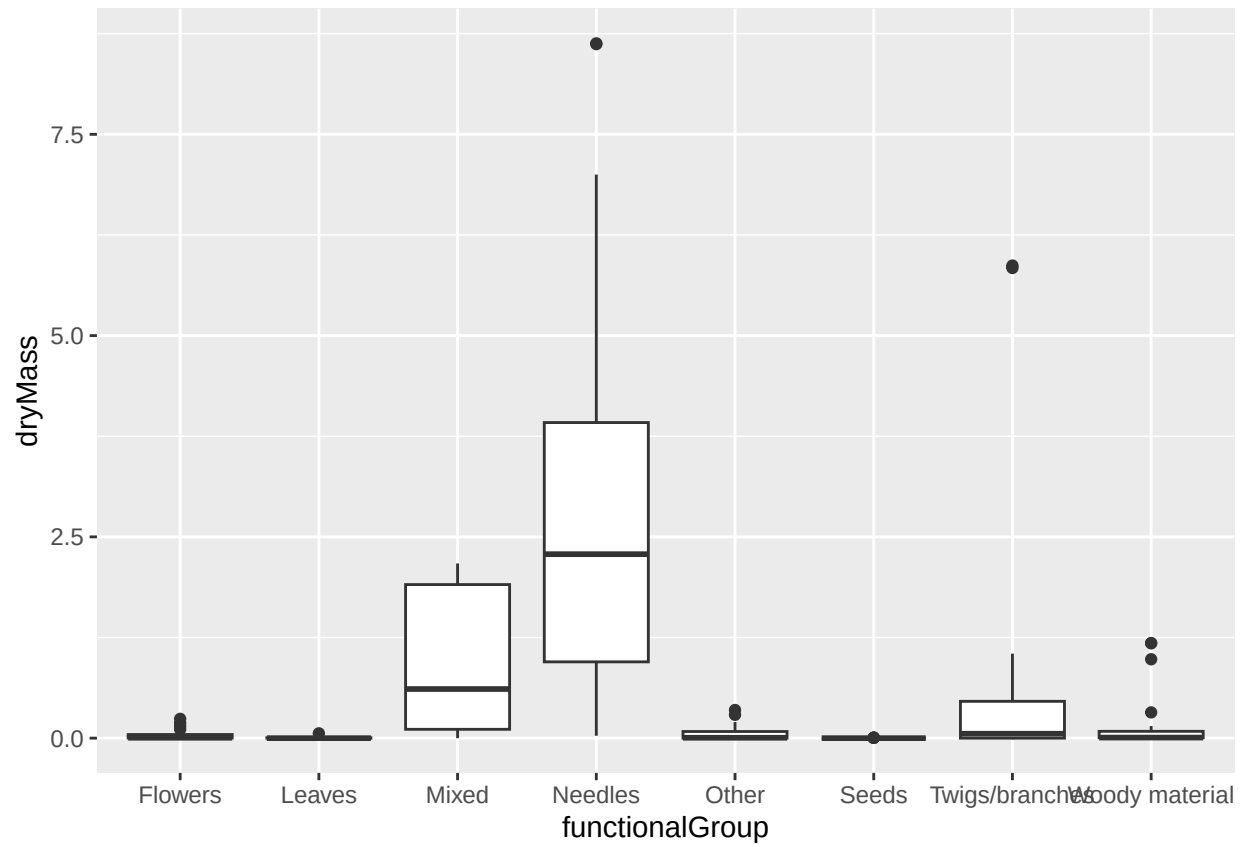
14. Create a bar graph of `functionalGroup` counts. This shows you what type of litter is collected at the Niwot Ridge sites. Notice that litter types are fairly equally distributed across the Niwot Ridge sites.

```
#Graph Functional Groups  
#Use ggplot to see how the count of records for each functional group  
ggplot(Litter, aes(x=functionalGroup)) +  
  geom_bar()
```

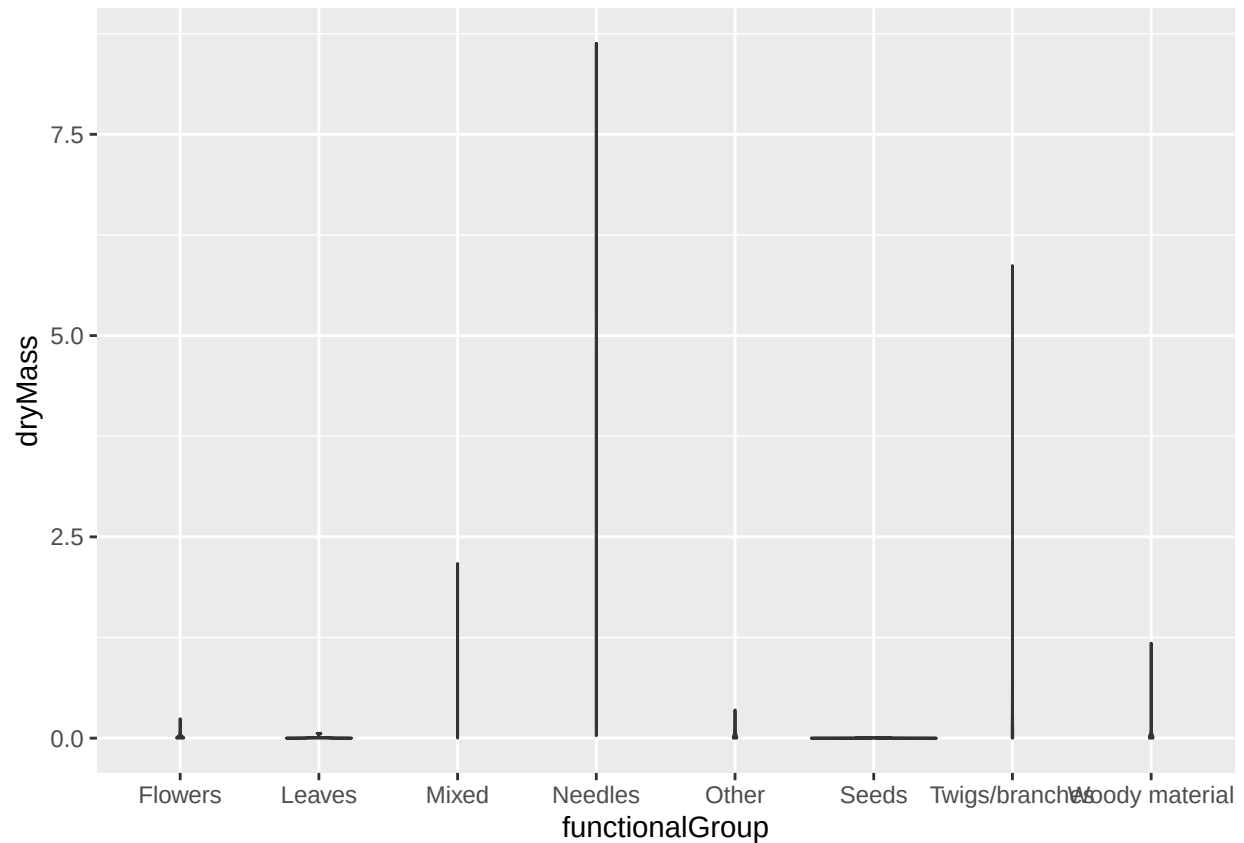


15. Using `geom_boxplot` and `geom_violin`, create a boxplot and a violin plot of `dryMass` by `functionalGroup`.

```
#Plot Dry Mass and Functional Group
#First, we use ggplot to create a box plot
ggplot(Litter, aes(x=functionalGroup, y=dryMass)) +
  geom_boxplot()
```

```
#Second, use ggplot to create violin plot
ggplot(Litter, aes(x=functionalGroup, y=dryMass)) +
  geom_violin()
```



Why is the boxplot a more effective visualization option than the violin plot in this case?

Answer: Violin plots are helpful for showing density, but in this case we are only interested in the range of values, so the boxplot is a more helpful visual to see the quartiles and the data range.

What type(s) of litter tend to have the highest biomass at these sites?

Answer: Needles are shown to have the highest biomass. This is evident as the mean falls above the interquartile ranges for all other litter types.